NEXUS MICRO-ESCROW PLATFORM

Institutional Production System Architecture Specification

Document Version: 1.0.0	**Target Environment:** Production Solo Dev
Core Gateway: Chapa Payment API	**Platform Anchor:** Telegram Web Apps (TWA)
Storage Layer: PostgreSQL (Strict ACID)	**State Machine:** Asynchronous Event-Driven

1. Executive Technical Overview

Nexus runs on a **Zero-Balance Holding Protocol**. Funds never sit in your bank account, transforming legal capital liabilities into API routing loops by offloading escrow custody entirely onto Chapa's central-bank-licensed settlement pools.

2. Core Architectural Topography

Layer Component	Tech/Framework	Structural Role & Security Boundary
Client Interface	TWA SDK + React 19	Executes directly inside native Telegram runtime. Implements state isolation and cryptographic Ir
Ingress Routing	Nginx + SSL TLS v1.3	Enforces rigid SSL termination, rate-limiting (max 60 req/min), and malicious payload blocking.
Application Server	Python 3.11 + FastAPI	Stateless execution layer. Implements strict dependency injection and HMAC webhook processo
Async Queue	Redis + Celery Task Queue	Handles the 48-hour automated escrow release timers, dispute notifications, and retry policies.
Data Vault	PostgreSQL 15 (Managed)	Single source of truth. Enforces atomic ACID transactions, multi-column uniqueness, and row loc

3. Transaction Ledger & State Machine

State Name	Transition Trigger	Operational Meaning & Asset Control Logic
DRAFT	User clicks 'Buy via Escrow'	Escrow metadata created. Awaiting buyer payment execution parameters.
PAYMENT_PENDING	Chapa API initialized	Transaction locked. Redirecting client to Chapa native portals.
HELD_IN_ESCROW	Validated Chapa Webhook	Funds cleared. Settled safely inside Chapa partner escrow pool.
SHIPPED	Seller uploads tracking	Logistics clock initialized. Start of 48-hour dispute countdown.
DISBURSED	Buyer approve / Timer expiry	Terminal state. API split execution triggered. Funds released to merchant.
DISPUTED	Buyer flags order in TWA	Escrow frozen. Transition to human arbitration layer. Portals open.

4. Production Database Architecture (PostgreSQL)

```
CREATE TYPE escrow_status AS ENUM ('DRAFT', 'PAYMENT_PENDING', 'HELD_IN_ESCROW', 'SHIPPED', 'DISBURSED', 'DIS
PUTED');

CREATE TABLE escrows (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    txn_reference VARCHAR(64) UNIQUE NOT NULL,
    buyer_telegram_id BIGINT NOT NULL,
    seller_telegram_id BIGINT NOT NULL,
    gross_amount_etb NUMERIC(12, 2) NOT NULL CHECK (gross_amount_etb >= 50.00),
    chapa_fee_etb NUMERIC(12, 2) NOT NULL,
    nexus_fee_etb NUMERIC(12, 2) NOT NULL,
    status escrow_status NOT NULL DEFAULT 'DRAFT',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);
```

5. Cryptographic Verification & Webhook Ingestion

```
import hmac, hashlib
from fastapi import Request, Header, HTTPException

async def verify_chapa_webhook(request: Request, chapa_signature: str = Header(...)):
    payload = await request.body()
    secret_key = os.environ.get("CHAPA_WEBHOOK_SECRET").encode('utf-8')
    computed_signature = hmac.new(secret_key, payload, hashlib.sha256).hexdigest()

    if not hmac.compare_digest(computed_signature, chapa_signature):
        raise HTTPException(status_code=401, detail="Signature Verification Mismatch")
    return True
```